



Final Exam

International Campus, MD Program

Cardiovascular System

February 2022 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Ranjbaran

Date of Exam: May 21, 2022

Duration of Exam: 100 Minutes

MCQ Question	89
Short Answer Questions	9

Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.
- For short answer Questions, write down your answer in the question pamphlet



Physiology: Dr. Ranjbaran

	Circulatory System Physiology	Question time (seconds): ➔	
1	Which type of vessel has zero mmHg pressure?		
	1	Vena Cava	
	2	Pulmonary artery	
	3	Arterioles	
	4	Aorta	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): ➔	
2	Flow (Q) through a blood vessel is determined by:		
	1	pressure difference and resistance of vessel	
	2	pressure difference	
	3	resistance of vessel	
	4	stroke volume	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): ➔	
3	Based on Poiseuille's equation, increasing radius of vessels		
	1	decrease flow of vessel	
	2	has no role on flow of vessel	
	3	increase length of vessel	
	4	increase flow of vessel	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): ➔	
4	How would a decrease in vascular resistance affect blood flow?		
	1	Increase at first and then decrease	
	2	Decrease blood flow	
	3	Increase blood flow	
	4	Has no role	



Source:	
Description:	

	Circulatory System Physiology	Question time (seconds): →	
5	The difference between systolic and diastolic pressure is called the		
	1	stroke volume	
	2	pulse pressure	
	3	compliance	
	4	resistance	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): →	
6	If a patient's systolic pressure is 90 mmHg and diastolic pressure is 60 mmHg, what's the mean arterial pressure?		
	1	100 mmHg	
	2	90	
	3	80	
	4	70	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): →	
7	Net filtration in capillary arteriole end is around:		
	1	13	
	2	7	
	3	33	
	4	20	
Source:			
Description:			

$$\frac{1}{3} \times S + \frac{2}{3} \times D$$

11



	Circulatory System Physiology	Question time (seconds): ➔	
8	Net reabsorption in capillary venule end is around:		
	1	13	
	2	7	
	3	33	
	4	20	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): ➔	
9	Capillary reabsorption is higher at:		
	1	arterial end of capillary	
	2	venous end of capillary	
	3	middle of capillary	
	4	metarteriole of capillary	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): ➔	
10	Which enzyme convert angiotensin I into angiotensin II?		
	1	Angiotensinogen	
	2	Angiotensin converting enzyme	
	3	Rrenin	
	4	Cortisol	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): ➔	
11	Which disease might increase cardiac output?		
	1	Hypothyroidism	
	2	Traumatic shock	
	3	Hyperthyroidism	
	4	Cardiac ischemia	
Source:			
Description:			



	Circulatory System Physiology	Question time (seconds): ➔	
12	Angiotensin II act directly on ... hormone and increase arterial pressure.		
	1	Endorphin	
	2	TSH	
	3	Testosterone	
	4	Aldosterone	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): ➔	
13	Which sentence about “Venous Return” is correct?		
	1	Is equal to pulse pressure.	
	2	Is detected in the capillaries.	
	3	Has no role on cardiac output.	
	4	Amount of blood flowing from the vein into the right atrium.	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): ➔	
14	Which organ is not considered as blood reservoir?		
	1	Brain	
	2	Spleen	
	3	Liver	
	4	Veins	
Source:			
Description:			

	Circulatory System Physiology	Question time (seconds): ➔	
15	Which protein has more effective role on plasma colloid osmotic pressure?		
	1	Transferrin	
	2	Creatine	
	3	Fibrinogen	
	4	Albumin	
Source:			
Description:			



	Circulatory System Physiology	Question time (seconds): ➔	
16	Which autonomic nervous system has more role on control of circulation?		
	1	Parasympathetic	
	2	Sympathetic	
	3	Parasympathetic and sympathetic	
	4	Brain stem	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
17	The followings cause the first rapid repolarization of cardiac ventricle EXCEPT:		
	1	rapid inactivation of I_{Na}	
	2	transient efflux of K^+	
	3	transient influx of Cl^-	
	4	rapid influx of Ca^{+2}	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
18	The duration of the plateau in atria is substantially less than in ventricular myocytes because		
	1	the magnitude of i_{to} during plateau is greater in atria than in ventricular myocytes	
	2	the magnitude of i_{to} during plateau is the same in atria and ventricular myocytes	
	3	the magnitude of i_{to} during plateau is less in atria than the ventricular myocytes	
	4	none of the above	
Source:			
Description:			



	Heart Physiology	Question time (seconds): ➔	
19	The resting cardiac cell membrane:		
	1	is primarily permeable to K ⁺ ions.	
	2	has resting membrane potential - 65 mV.	
	3	favors inward movement of K ⁺ ions at -85 mV.	
	4	influenced by extra-cellular Na ⁺ ion concentration.	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
20	The transmembrane potential during phase 4 is much less negative in nodal cells than in atrial and ventricular myocytes because		
	1	nodal cells lack the i _{K1} type of K ⁺ channel	
	2	nodal cells have high i _{K1} type of K ⁺ channel	
	3	nodal cells lack the funny Na ⁺ channel	
	4	nodal cells have high funny Na ⁺ channel	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
21	Cardiac cells:		
	1	represent 90 % of the total volume of the heart.	
	2	are rich in mitochondria (30% of cell volume) as in skeletal muscle.	
	3	contain sarcoplasmic reticulum which takes up Ca ²⁺ during contraction cycle.	
	4	function as a syncytium.	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
22	Regarding the afterload in the whole intact heart:		
	1	is the resistance against which blood is expelled.	
	2	it equals the end systolic pressure	
	3	approximately equals the aortic pressure.	
	4	all of the above.	



Source:	
Description:	

	Heart Physiology		Question time (seconds): ➔	
23	Ventricular end-diastolic pressure can be used instead of the usual variable on the abscissa (x-axis) when plotting a Frank-Starling curve. What variable does the x-axis usually represent on a Frank-Starling curve?			
	1	cardiac output		
	2	ventricular stroke volume		
	3	ventricular end-diastolic volume		
	4	cardiac work		
Source:				
Description:				

	Heart Physiology		Question time (seconds): ➔	
24	During isovolumetric ventricular contraction phase:			
	1	mitral and tricuspid valves close causing second heart sound.		
	2	intraventricular pressure increases without change in ventricular volume.		
	3	atrioventricular valves bulge into the atria causing a drop in atrial pressure.		
	4	QRS complex coincides with ventricular contraction.		
Source:				
Description:				

	Heart Physiology	Question time (seconds): ➔	
25	Regarding atrial systole all the followings occurs EXCEPT:		
	1	mitral and tricuspid valves are open and the aortic and pulmonary valves are closed.	
	2	contraction of the atria pushes 30% of blood into the ventricles.	
	3	contraction of the atrial muscle prevents regurgitation of blood into the veins.	
	4	P wave precedes atrial systole.	
Source:			
Description:			



	Heart Physiology	Question time (seconds): ➔	
26	A hospitalized patient has an ejection fraction of 0.4, a heart rate of 95 beats/min, and cardiac output of 3.5 L/min. what is the patient's end-diastolic volume?		
	1	14 ml	
	2	37 ml	
	3	55 ml	
	4	92 ml	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
27	In aortic pressure curve:		
	1	during ventricular systole walls of the aorta and arteries are stretched causing systolic pressure	
	2	dicrotic notch occurs when the aortic valve opens.	
	3	incisura is caused by forward flow of blood before closure of the valve.	
	4	elastic recoil of walls of the aorta and arteries maintain diastolic pressure.	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
28	During cardiac cycle the atrial pressure:		
	1	rises during atrial systole and drops during isovolumetric contraction.	
	2	falls rapidly during ventricular ejection.	
	3	rises as blood flows passively into the ventricles.	
	4	decreases markedly in tricuspid regurgitation.	
Source:			
Description:			



	Heart Physiology	Question time (seconds): ➔	
29	The v wave in the atrial pressure curve that occurs toward the end of ventricular contraction is due to		
	1	atrial contraction	
	2	bulging of the A-V valves backward toward atria	
	3	slow flow of blood into atria from veins while A-V valves are closed	
	4	slight backflow of blood into the atria at the onset of ventricular contraction	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
30	High tension-time index in the cardiac muscle cells occur during		
	1	isovolumetric contraction	
	2	normal left ventricular diameter	
	3	abnormally dilated left ventricle	
	4	isovolumetric relaxation	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
31	A decrease in atrioventricular nodal conduction velocity will		
	1	decrease heart rate	
	2	increase P wave amplitude	
	3	increase PR interval	
	4	widen the QRS complex	
Source:			
Description:			

	Heart Physiology	Question time (seconds): ➔	
32	The ECG findings of bundle branch block (BBB):		
	1	associated with absence of P waves.	
	2	heart rate is slowed <60 b/min.	
	3	double hump QRS complex occurs in V5 & V6 in right BBB.	
	4	QRS complexes are prolonged & deformed.	



Source:	
Description:	

	Heart Physiology		Question time (seconds): ➔	
33	In a standard lead III electrocardiography, the negative terminal is connected to			
	1	the left arm		
	2	the left leg		
	3	the right arm		
	4	the right leg		
Source:				
Description:				

	Blood Physiology		Question time (seconds): ➔	
34	Which of the following is a major determinant of red blood cells (RBCs) production rate?			
	1	The number of RBCs that should be regulated within narrow limits		
	2	Concentration of RBCs in the blood		
	3	Hematocrit rate		
	4	The amount of oxygen transported to the tissues in relation to tissue demand for oxygen		
Source:				
Description:				

	Blood Physiology		Question time (seconds): ➔
35	Which of the following describes best the biggest problem with suffers from hereditary spherocytosis?		
	1	When the patient is exposed to low concentrations of oxygen, the RBCs are unable to carry oxygen.	
	2	The RBCs that are formed are mostly oversized, have bizarre shapes, and have fragile membranes	
	3	Lack of functioning bone marrow	
	4	The RBCs cannot withstand compression forces because they do not have the normal loose, baglike cell membrane structure of the biconcave discs	
Source:			
Description:			



	Blood Physiology	Question time (seconds): ➔	
36	Which of the following refers to the effects of anemia on circulatory system function?		
	1	Increased pumping workload	on the heart
	2	Increased the resistance to blood flow in the peripheral blood vessels	
	3	Decreased venous return	
	4	Increased blood pressure	
Source:			
Description:			

	Blood Physiology	Question time (seconds): ➔	
37	Which factor mediates renal tissue hypoxia that leads to increased erythropoietin gene expression?		
	1	Epinephrine and norepinephrine	
	2	Hypoxia-inducible factor-1	
	3	Prostaglandins	
	4	Interleukin-3	
Source:			
Description:			

	Blood Physiology	Question time (seconds): ➔	
38	which of the following refers to the rate-limiting factor in causing blood coagulation?		
	1	Polymerization of fibrinogen molecules into fibrin fibers	
	2	Formation of prothrombin activator	
	3	Conversion of prothrombin to thrombin	
	4	Ca concentration in blood	
Source:			
Description:			



	Blood Physiology	Question time (seconds): ➔	
39	All of the followings are used to prevent blood clotting outside the body <u>EXCEPT</u>:		
	1	blood collected in siliconized containers	
	2	heparin	
	3	citrate ions	
	4	warfarin	
Source:			
Description:			

	Blood Physiology	Question time (seconds): ➔	
40	One of the most prevalent causes of vitamin K deficiency is		
	1	the absence of vitamin K from the diet	
	2	liver carboxylase deficiency	
	3	failure of the liver to secrete bile into the gastrointestinal tract	
	4	splenectomy	
Source:			
Description:			

	Blood Physiology	Question time (seconds): ➔	
41	Which factor is <u>only</u> involved in the intrinsic pathway of blood coagulation?		
	1	X	
	2	VII	
	3	VI	
	4	XI	
Source:			
Description:			

	Blood Physiology	Question time (seconds): ➔	
42	Where is major basic protein secreted from and what role does it play?		
	1	Basophils, involvement in allergic reactions	
	2	Eosinophils, larvicidal property	
	3	Basophils, larvicidal property	
	4	Eosinophils, involvement in allergic reactions	



Source:	
Description:	

	Blood Physiology		Question time (seconds): ➔	
43	Which of the following refers to the difference between neutrophils and macrophages?			
	1	Lipase-rich lysosomes		
	2	Strong oxidizing agents		
	3	Phagocytosis of bacteria		
	4	Proteolytic enzymes		
Source:				
Description:				

	Blood Physiology		Question time (seconds): ➔
44	This type of polycythemia is caused by a genetic aberration in the hemocytoblastic cells in which the RBC count may be 7 to 8 million/mm3 and the hematocrit may be 60% to 70%.		
	1	Polycythemia Vera (Erythremia)	
	2	Secondary polycythemia	
	3	Primary polycythemia	
	4	Physiologic polycythemia	
Source:			
Description:			

	Blood Physiology		Question time (seconds):	
45	Which of the following plays an important role in the clot retraction?			
	1	Platelets		
	2	Fibrin fibers		
	3	von Willebrand factor		
	4	Platelet-activating factor		
Source:				
Description:				



Anatomy: Dr. Pasbakhsh

	Anatomy of cardiovascular block		Question time (seconds): ➔	
46	Which of the following muscles is in the deep layer of the erector spin muscle?			
	1	Longissimus thoracis		
	2	Multifidus		
	3	Iliocostalis		
	4	Serratus posterior inferior		
Source:				
Description:				

	Anatomy of cardiovascular block		Question time (seconds): ➔	
47	Which of the following structures is more to the right of the body than the rest?			
	1	Right ventericle		
	2	Right Vagus nerve		
	3	Right Phrenic nerve		
	4	Pericard		
Source:				
Description:				

	Anatomy of cardiovascular block		Question time (seconds): ➔	
48	The right margin of heart is formed mostly by which champer of the heart?			
	1	Left atrium		
	2	Right atrium		
	3	Left ventricle		
	4	Right ventricle		
Source:				
Description:				



	Anatomy of cardiovascular block	Question time (seconds): →	
49	Which of the following statement is correct?		
	1	8 th Anterior intercostal artery arises from Musculophrenic artery	
	2	Thoracic aorta pass posterior to right atrium	
	3	Subcostal artery arises from internal thoracic artery	
	4	Left and right internal thoracic veins drains to Azygous vein	
Source:			
Description:			

	Anatomy of cardiovascular block	Question time (seconds): →	
50	All of the following structures are visible posterior to thoracic aorta except		
	1	Right posterior intercostal artery	
	2	Accessory hemiazygous vein	
	3	Hemi Azygous vein	
	4	Thoracic vertebra	
Source:			
Description:			

	Anatomy of cardiovascular block	Question time (seconds): →	
51	In the case of the intervertebral disc,all of the following statements are true except		
	1	Annulus fibrosis are located in the peripheral part of the disc	
	2	Disc herniation usually occurs in the antero lateral direction	
	3	Anterior portion of Annulus fibrosis is thicker than posterior part	
	4	Intervertebral disc is composed of the nucleus pulposus and Annulus fibrosis	
Source:			
Description:			



	Anatomy of cardiovascular block	Question time (seconds): →	
52	Which of the following structures are going to innervate by phrenic nerve?		
	1	Diaphragm	
	2	Visceral serous pericard	
	3	Pectoralis minor	
	4	Pectoralis major	
Source:			
Description:			

	Anatomy of cardiovascular block	Question time (seconds): →	
53	Which of the following structure is found both inferior and posterior the arch of the aorta?		
	1	Left recurrent laryngeal nerve	
	2	Left vagus nerve	
	3	Deep cardiac plexus	
	4	Superficial cardiacplexus	
Source:			
Description:			

	Anatomy of cardiovascular block	Question time (seconds): →	
54	During a heart procedure the surgeon accidentally damages the vein that is accompanied by the posterior interventricular artery, which of the following veins is most likely to be damaged?		
	1	Great cardiac vein	
	2	Middle cardiac vein	
	3	Small cardiac vein	
	4	Left marginal vein	
Source:			
Description:			



	Anatomy of cardiovascular block	Question time (seconds): →	
55	<i>Which of following ribs has a single articular facet on its head?</i>		
	1	Second rib	
	2	first rib	
	3	Seventh rib	
	4	Ninth rib	
Source:			
Description:			

	Anatomy of cardiovascular block	Question time (seconds): →	
56	Pathological mass is find in superior mediastinum, which of the following structures <u>may not</u> be compressed?		
	1	Right phrenic nerve	
	2	Left brachiocephalic vein	
	3	Right recurrent laryngeal nerve	
	4	Thoracic duct	
Source:			
Description:			

	Anatomy of cardiovascular block	Question time (seconds): →	
57	The third rib is going to articulate with all of the following except		
	1	Body of second thoracic vertebra	
	2	Body of third thoracic vertebra	
	3	Transvers process of second thoracic vertebra	
	4	Disk between second and third vertebra	
Source:			
Description:			



	Anatomy of cardiovascular block		Question time (seconds): →	
58	Which of the following artery arises from internal thoracic artery?			
	1	Musculophrenic artery		
	2	Superior phrenic		
	3	Subcostal		
	4	Common carotid		
Source:				
Description:				

	Anatomy of cardiovascular block		Question time (seconds): →	
59	Where is sinu-atrial node?			
	1	In right atrium		
	2	In left atrium		
	3	In right ventricle		
	4	In left ventricle		
Source:				
Description:				

	Anatomy of cardiovascular block		Question time (seconds): →	
60	Where the deep cardiac plexus has seen?			
	1	In posterior mediastinum		
	2	Anterior to bifurcation of Aorta		
	3	Posterior to arch of Aorta		
	4	In anterior mediastinum		
Source:				
Description:				

	Anatomy of cardiovascular block		Question time (seconds): →	
61	Where we should listen to Mitral valve sounds?			
	1	In left 5th intercostal space over apex of heart		
	2	In 6 th right intercostal space		
	3	In 2th left intercostal space		
	4	In 2th right intercostal space		



Source:	
Description:	

	Anatomy of cardiovascular block		Question time (seconds): ➔	
62	All of following joints are the type of synovial joints except?			
	1	First Costo Vertebral		
	2	First Costo Transverse		
	3	First Costo Sternal		
	4	Second Costo Sternal		
Source:				
Description:				

	Anatomy of cardiovascular block		Question time (seconds): ➔	
63	The 5th posterior intercostal arteries on each side are derived from which artery			
	1	Subclavian		
	2	Left Internal thoracic		
	3	Right internal thoracic		
	4	Thoracic aorta		
Source:				
Description:				

	Anatomy of cardiovascular block		Question time (seconds): ➔	
64	All of the following muscles are innervated by the anterior branch (ventral) of the intercostal nerves except			
	1	Iliocostalis thoracic		
	2	Serratus posterior inferior		
	3	Serratus posterior superior		
	4	Transverse thoracic		
Source:				
Description:				



	Anatomy of cardiovascular block		Question time (seconds): ➔	
65	In the case of the sympathetic chain, all of the following definitions are correct except?			
	1	The sympathetic chain is also seen in the abdomen		
	2	Greater splanchnic nerves contain myelin		
	3	White rami is found in lateral side to gray rami		
	4	White rami is along the entire length of the sympathetic chain		
Source:				
Description:				

	Anatomy of cardiovascular block		Question time (seconds): ➔	
66	Lymphatic drainage of all the following areas into the thoracic duct except?..			
	1	Right lower limb		
	2	Left lower limb		
	3	Right posterior second intercostal space		
	4	Left posterior first intercostal space		
Source:				
Description:				

	Anatomy of cardiovascular block		Question time (seconds): ➔	
67	All the following muscles are connected to the third rib except			
	1	Pectoralis major		
	2	Pectoralis minor		
	3	Trapezius		
	4	Serratus posterior superior		
Source:				
Description:				



	Anatomy of cardiovascular block	Question time (seconds): ➔	
68	All the following options are correct except?...		
	1	The thoracic aorta begins on the left side of the fourth thoracic vertebra	
	2	The pericardio co phrenic vessels accompany the phrenic nerve in the chest	
	3	The thymus is found in the posterior mediastinum	
	4	The left brachiocephalic vein is longer than the right	
Source:			
Description:			

	Anatomy of cardiovascular block	Question time (seconds): ➔	
69	All the following options are correct except....		
	1	The abdominal aorta divides into two branches around the fourth lumbar vertebra	
	2	The pericardio co phrenic artery drives from the Internal thoracic artery	
	3	The posterior intercostal arteries drives from the Internal thoracic artery	
	4	The left posterior intercostal artery is longer than the right	
Source:			
Description:			

	Anatomy of cardiovascular block	Question time (seconds): ➔	
70	All of the following veins open to right atrium except....		
	1	Inferior vena cava	
	2	Left pulmonary vein	
	3	Superior vena cava	
	4	Coronary sinus	
Source:			
Description:			



Histology: Dr. Malek

	Histology	Question time (seconds): ➔	
71	Which one is primary lymphoid organ?		
	1	Lymphnode	
	2	Spleen	
	3	Tonsil	
	4	Thymus	
Source:			
Description:			

	Histology	Question time (seconds): ➔	
72	Where we observe Hassall corpuscle in thymus?		
	1	Cortex of lobule	
	2	Medulla of lobule	
	3	Capsule	
	4	Terabecula	
Source:			
Description:			



	Histology		Question time (seconds): ➔	
73	Which one is antibody of secretions such as saliva?			
	1	IgA 		
	2	IgG		
	3	IgM		
	4	IgD		
Source:				
Description:				

	Histology		Question time (seconds): ➔	
74	Where we observe T cell in lymphnodes?			
	1	Cortex		
	2	Medulla		
	3	Paracortex 		
	4	Cortical sinus		
Source:				
Description:				

	Histology		Question time (seconds): ➔	
75	which organ is site of blood filtration?			
	1	Spleen 		
	2	Thymus		
	3	Tonsil		
	4	Malt		
Source:				
Description:				



	Histology	Question time (seconds): →	
76	In which vessel we can find internal elastic layer?		
	1	Aorta	
	2	Venule	
	3	Capillary	
	4	Muscular artery	
Source:			
Description:			

	Histology	Question time (seconds): →	
77	Which one is the thickest layer of heart?		
	1	Myocardium	
	2	Endocardium	
	3	Pericardium	
	4	Epicardium	
Source:			
Description:			

	Histology	Question time (seconds): →	
78	What is the thickest layer of arteries?		
	1	Intima	
	2	Media	
	3	Adventitia in veins	
	4	Seroza	
Source:			
Description:			



	Histology	Question time (seconds): ➔	
79	In which vessel we observe valves?		
	1	Lymphatic vessels	
	2	Arteries	
	3	Aorta	
	4	Capillary	
Source:			
Description:			

	Histology	Question time (seconds): ➔	
80	Most numerous vessels in our body are ...		
	1	Small arteries	
	2	Veins	
	3	Capillaries	
	4	Large arteries	
Source:			
Description:			

Continue to next page



Embryology: Dr. Amidi

	Embryology	Question time (seconds): ➔	
81	which of the following form the floor fossa ovalis?		
	1	septum primum.	
	2	septum secundum.	
	3	sub endocardial cushions.	
	4	septum intermedium.	
Source:			
Description:			

	Embryology	Question time (seconds): ➔	
82	The left horn of sinus venosus forms:		
	1	The Rough Anterior wall of the right atrium.	
	2	The Smooth Posterior wall of the right atrium.	
	3	Atrophy and forms coronary sinus	
	4	The middle cardiac vein	
Source:			
Description:			

	Embryology	Question time (seconds): ➔	
83	Which of the follow is NOT part of TETRALOGY OF FALLOT:		
	1	Pulmonary stenosis	
	2	Overriding of the aorta	
	3	Thickened (Hypertrophy)right ventricle wall	
	4	ASD	
Source:			
Description:			



	Embryology	Question time (seconds): ➔	
84	During vascular development, the terminal part of IVC arises from:		
	1	Right umbilical veins.	
	2	Right vitelline veins.	
	3	Right subcardinal veins.	
	4	Right supracardinal veins.	
Source:			
Description:			

	Embryology	Question time (seconds): ➔	
85	During Aortic arch development, Fate of the following are correct EXCEPT:		
	1	Aortic sac: proximal segment of the aortic arch and brachiocephalic trunk.	
	2	1st aortic arch: maxillary artery (in the face)	
	3	2nd aortic arch: stapedial artery (in middle ear)	
	4	3rd aortic arch: common carotid and initial segments of external carotid artery	
Source:			
Description:			

	Embryology	Question time (seconds): ➔	
86	Failure of the aorticopulmonary septum formation will result in		
	1	Tetralogy of Fallot	
	2	Transposition of the great vessels	
	3	Persistent truncus arteriosus	
	4	An atrial-septal defect	
Source:			
Description:			



	Embryology	Question time (seconds): ➔	
87	Which part of the primitive heart tube gives rise to outflow right ventricle ?		
	1	Bulbus cordis	
	2	Primitive atrium	
	3	Primitive ventricle	
	4	Sinus venosus	
Source:			
Description:			

	Embryology	Question time (seconds): ➔	
88	The embryo cardinal venous vessels which drain into the sinus venosus are:		
	1	anterior cardinal veins	
	2	common cardinal veins	
	3	inferior cardinal veins	
	4	superior cardinal veins	
Source:			
Description:			

	Embryology	Question time (seconds): ➔	
89	Which of the following statements concerning the aorticopulmonary septum is <u>INCORRECT</u> ?		
	1	The aorticopulmonary septum partitions the outflow channel of the ventricles into the aorta and the pulmonary trunk.	
	2	The aorticopulmonary septum is largely derived from neural crest cells	
	3	The aorticopulmonary septum contributes to the formation of the muscular interventricular septum	
	4	The aorticopulmonary septum undergoes a 180o spiral	
Source:			
Description:			

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